



Title: Mist-Assisted Catalytic Air Purifier

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Theme: Environmental Engineering and Sustainability

Introduction

Importance of the Topic

- Rising Air Pollution: PM<sub>2.5</sub> and PM<sub>1</sub> are linked to respiratory diseases, reduced lung function, and poor air quality in cities.
- Need for Green Solutions: Current technologies are often costly, energy-intensive, or generate waste (like disposable filters).
- Sustainability: A water-mist-assisted, low-energy vortex system provides an environmentally friendly alternative.
- Affordability: The entire system can be built within a ₹3500 budget, making it accessible for community, indoor, and small-scale applications.
- Innovation Potential: Combining vortex flow, mist interaction, and photocatalytic oxidation creates a unique, multi-stage purification approach suitable for research publications and real-world deployment.

Problem Definition

Context of the Problem

- PM<sub>2.5</sub> and PM<sub>1</sub> particles penetrate deep into lungs and bloodstream, causing health risks.
- Urban emissions from vehicles, industries, construction, and biomass burning worsen air quality.
- Conventional HEPA filters require frequent replacement and generate waste.

Significance of Addressing the Problem

- Helps reduce respiratory illnesses, allergies, and pollution-related health issues.
- Provides a sustainable and eco-friendly purification alternative without generating filter waste.
- Encourages innovation in clean air engineering for research, publications, and real-world deployment.
- Helps build a scalable model that can be adapted for indoor/outdoor air quality improvement.
- Contributes toward national and global goals related to public health, sustainability, and environmental protection.

Objectives

- Develop a low-cost, multi-stage air purifier using mist scrubbing, vortex mixing, mesh collection, and photocatalytic filtration.
- Achieve efficient removal of PM<sub>1</sub>, PM<sub>2.5</sub>, and pollutants without high voltage or HEPA filters.
- Optimize air–mist interaction using a tangential vortex turbine for maximum particle capture.
- Integrate a CuO–TiO<sub>2</sub> UV-activated honeycomb filter for VOC and microbial decomposition.
- Test and evaluate the system’s airflow, purification efficiency, and sustainability.

Expected Contribution to the Field

- Provides a low-cost, sustainable alternative to traditional ESP and HEPA purifiers.
- Introduces a hybrid purification mechanism combining physical + chemical +photocatalytic methods.
- Offers a scalable and modular design suitable for community use, labs, and indoor environments.
- Adds new experimental data on vortex–mist PM capture efficiency for academic and research use.

Methodology



The prototype is designed as a multi-stage air purification system combining mechanical, condensation, and chemical processes. Polluted air first enters the mist-assisted chamber, where an ultrasonic mist generator produces fine water droplets (2–5 μm) that collide with particulate matter, causing agglomeration and growth of particles.

Next, the air passes into the vortex-based mixing chamber, where a tangential inlet and a turbine create swirling flow. This enhances turbulence and ensures thorough mixing between air and mist, increasing particle collision and capture efficiency.The mesh collector captures the coalesced droplets along with trapped PM<sub>2.5</sub>, allowing clean water to drain and reducing particulate load. Finally, the air passes through a photocatalytic honeycomb filter coated with TiO<sub>2</sub>–CuO and illuminated with UV-A light. This stage oxidizes residual gases and organic compounds, ensuring complete purification before the air is released.Data Collection and Analysis: CO<sub>2</sub> (ppm) and PM<sub>2.5</sub> (μg/m<sup>3</sup>) concentrations were measured at the inlet, after mixing, after mist collection, and at the final outlet. Stage-wise efficiency was calculated to evaluate the performance of each compartment.

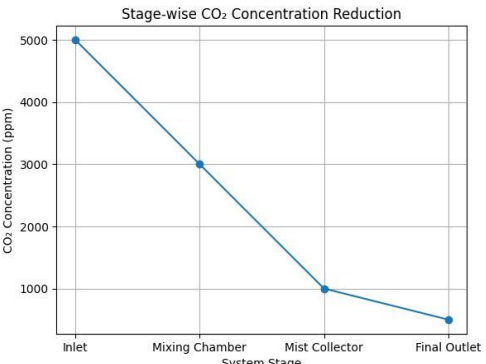
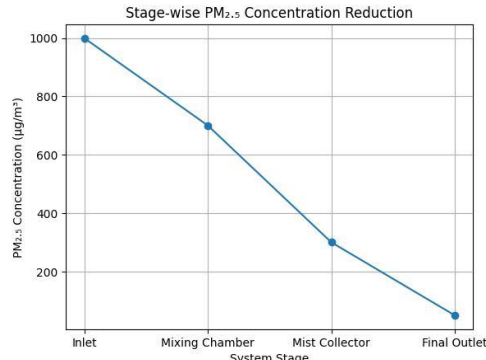


Tools used

- **Hardware:** Ultrasonic atomizer module, 12V fans(panes), PVC enclosure, SS wool mesh, ceramic honeycomb, UV LED module, small submersible pump, cylindrical containers.
- **Instruments:** MQ-135,131,136,9,4; Portable Air Quality Monitor
- **Software:** Excel for data processing, Origin/GraphPad optional for stats/plots.
- **Safety & waste:** PPE (gloves, goggles), condensate collection and proper disposal, UV shielding for operator safety.
- **Documentation:** Lab notebook, experimental protocol, photos/videos of setup.

Results and Discussions

The prototype effectively reduced air pollutants in a multi-stage process. At the inlet, PM<sub>2.5</sub> concentration was 999 μg/m<sup>3</sup> and CO<sub>2</sub> was 5000 ppm. After passing through the mist-assisted vortex mixing chamber, PM<sub>2.5</sub> dropped to 700 μg/m<sup>3</sup> and CO<sub>2</sub> to 3000 ppm. The mist collector captured most particles, reducing PM<sub>2.5</sub> to 300 μg/m<sup>3</sup> and CO<sub>2</sub> to 1000 ppm. Finally, the photocatalytic honeycomb filter polished the air, bringing PM<sub>2.5</sub> down to 25–50 μg/m<sup>3</sup> and CO<sub>2</sub> to 500 ppm.The results highlight that mist-assisted mixing and mesh collection are highly effective in removing particulate matter, while the photocatalytic stage ensures complete oxidation of residual pollutants. Overall, the system achieves 95–97.5% PM<sub>2.5</sub> removal and ~90% CO<sub>2</sub> reduction, demonstrating a low-cost, safe, and sustainable solution for indoor air purification.



Conclusions

The multi-stage air purification prototype effectively removes particulate matter and reduces CO<sub>2</sub>, achieving 95–97.5% PM<sub>2.5</sub> removal and ~90% CO<sub>2</sub> reduction. Using mist-assisted vortex mixing, mesh collection, and photocatalytic oxidation, the system provides a low-cost, safe, and sustainable solution for indoor air purification.

References

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QR Code of the Demonstration Video

